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Subject Code: BMC053

Roll No:

MCA

(SEM. IV) THEORY EXAMINATION 2025-26

NATURAL LANGUAGE PROCESSING

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

(a)	Define Natural Language Processing. Mention any two goals of NLP.	CO1	K2
(b)	List any four levels of language analysis used in Natural Language Understanding.	CO1	K2
(c)	State any two real-world applications of NLP and mention their purpose.	CO2	K4
(d)	What is parsing in NLP? Name any two types of parsers.	CO3	K3
(e)	Define context-free grammar. Give one example of a grammar rule.	CO4	K2
(f)	What is Part-of-Speech tagging? Why is it required in NLP?	CO5	K3
(g)	Define lexical ambiguity with a suitable example.	CO5	K3

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	Explain different levels of language analysis in NLP. Discuss how these levels help in building Natural Language Understanding systems.	CO1	K2
(b)	Analyze the role of NLP in machine translation and database interface applications with suitable examples.	CO2	K4
(c)	Explain top-down and bottom-up parsing techniques. Compare their working with a suitable sentence example.	CO3	K3
(d)	Discuss feature systems in English grammar. Explain how morphological analysis and lexicon support parsing with features.	CO4	K3
(e)	Explain probabilistic language processing. Discuss how lexical probabilities are obtained and used in Part-of-Speech tagging.	CO5	K5

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Describe the organization of a Natural Language Understanding system. Explain the role of representation and understanding in NLP.	CO1	K2
(b)	Explain the outline of English syntax. Discuss the importance of noun phrases, verb phrases and sentence structure in NLP.	CO1	K2

4. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain semantics and knowledge representation in NLP. Discuss their importance in language understanding systems.	CO2	K4
(b)	Analyze the use of NLP in database interface systems. Explain how natural language queries are converted into structured database queries.	CO2	K4

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain grammar and sentence structure in NLP. Describe top-down chart parsing with a suitable example.	CO3	K3
(b)	Describe Transition Network Grammars and Augmented Transition Networks. Explain their use in parsing natural language sentences.	CO3	K3

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain auxiliary verbs and verb phrases in natural language grammar. Discuss their role in sentence interpretation.	CO4	K3
(b)	Describe the movement phenomenon in language. Explain how questions can be handled using context-free grammars.	CO4	K3

7. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain Probabilistic Context-Free Grammar. Discuss how it helps in handling and evaluating syntactic ambiguity.	CO5	K5
(b)	Explain word senses and ambiguity. Describe how ambiguity can be encoded in logical form with a suitable example.	CO5	K5